Adaptive Approximation

Adaptive Finite Element Methods for Elliptic PDEs — Lecture 4

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Why adapt?

- **Limitation of uniform meshes:** On a quasi-uniform mesh the convergence rate is capped by the global L^2-based Sobolev regularity of u.
- **Adaptive remedy:** Adaptivity instead tailors the mesh to u through a nonlinear process and measures the regularity in an L^p-based Sobolev space with $p < 2$.

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The plan of this lecture

- **Why:** error equidistribution — a 1-D graded-mesh model, then the 2-D heuristic; the singular class W^2_1 reaches the same rate as H^2.
- **How:** *bisection* (newest-vertex bisection) builds graded, conforming, shape-regular meshes by purely local operations.
- **Cost:** the Binev–Dahmen–DeVore / Stevenson complexity bound $\#\mathscr{T}_k - \#\mathscr{T}_0 \lesssim \sum_j \#\mathscr{M}_j$.
- **Optimality:** a *greedy* algorithm; W^2_p regularity $(p > 1)$ yields the optimal rate $N^{-1/2}$.

Nonlinear approximation: the mesh is designed for the particular u at hand.

Roadmap

Why adapt? Error equidistribution

One-dimensional model problem: uniform mesh

- **Setting:** Let $\Omega = (0,1)$ and approximate an absolutely continuous v by a piecewise-constant v_N on a mesh with N intervals $I_i = [x_{i-1}, x_i]$.

- **Uniform mesh - smooth data:**

If $v \in W^1_\infty(\Omega)$ and $\mathscr{T}$ is uniform ($h_i = h = 1/N$), integrating $|v'|$ over I_i gives

$$|v(x) - v_N(x)| \le h\,\|v'\|_{L^\infty(\Omega)} \quad \Rightarrow \quad \|v - v_N\|_{L^\infty(\Omega)} \le \frac{1}{N}\,|v|_{W^1_\infty(\Omega)}.$$

- **Remarks:**
  - The rate N^{-1} requires the strong regularity $v \in W^1_\infty$.
  - What if v' is unbounded (a singularity)? A uniform mesh no longer delivers this rate.
  - If $v \in C^{0,\gamma}$ for $0 < \gamma < 1$, then $\|v - v_N\|_{L^\infty(\Omega)} \lesssim N^{-\gamma}$.

One-dimensional model problem: Graded mesh

- **Rough function:** Let $v \in W_1^1(\Omega)$ only, possibly with unbounded derivative. Introduce the monotone error-accumulation function

$$\phi(x) = \int_0^x |v'(s)| \, \mathrm{d}s, \qquad \phi(0) = 0, \quad \phi(1) = |v|_{W_1^1(\Omega)} \, .$$

- **Grade by equidistributing the *range* of ϕ:**

Choose nodes x_i so that

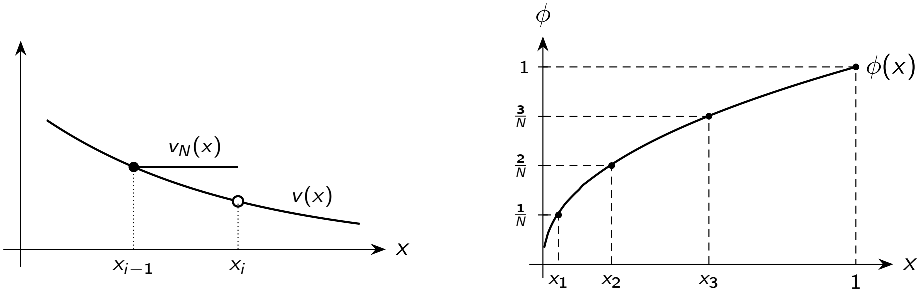
$$\phi(x_i) = \frac{i}{N} \, |v|_{W_1^1(\Omega)} \qquad (0 \le i \le N),$$

i.e. cut the range of ϕ into N equal pieces.

- **Error equidistribution:** For $x \in I_i$ this equalizes the local errors:

$$|v(x) - v_N(x)| \le \int_{x_{i-1}}^{x_i} |v'| \, \mathrm{d}s = \phi(x_i) - \phi(x_{i-1}) = \frac{1}{N} \, |v|_{W_1^1(\Omega)} \, , \quad \Rightarrow \quad \|v - v_N\|_{L^\infty(\Omega)} \le \frac{1}{N} \, |v|_{W_1^1(\Omega)} \, .$$

The 1-D mechanism for error equidistribution



(a) piecewise-constant v_N (b) equidistributing the range of ϕ

Nodes x_i (via $\phi(x_i) = i/N$) make every subinterval carry the same error $1/N$; they cluster where $|v'|$ is large.

The singular example $v(x) = x^\gamma$

- **A singular test function:** Take $v(x) = x^\gamma$, $0 < \gamma < 1$. Then

$$v'(x) = \gamma x^{\gamma-1} \notin L^\infty(0,1),$$

so uniform mesh gives $\|v - v_N\|_{L^\infty(\Omega)} \lesssim h^\gamma$; yet $v \in W_1^1(\Omega)$ and

$$\phi(x) = \int_0^x \gamma s^{\gamma-1}\,\mathrm{d}s = x^\gamma = v(x).$$

- **Error accumulation:** The error-accumulation function is the solution itself: its singular behavior at the origin is exactly what makes a graded mesh profitable.
- **Sobolev numbers:** The graded mesh recovers the *same* $O(1/N)$ rate for the rougher class W_1^1 that the uniform mesh gives for W_∞^1. This hinges on equality of the *Sobolev numbers*:

$$\mathrm{sob}(W_1^1) = 1 - \tfrac{1}{1} \ \geq \ 0 - \tfrac{1}{\infty} = \mathrm{sob}(L^\infty).$$

Heuristics in two dimensions

- **Local interpolation estimate:** The starting point is a local interpolation estimate carrying a vanishing power of h_T. One has $W_1^2(\Omega) \hookrightarrow C^0(\bar{\Omega})$, so $I_{\mathscr{T}} u$ is well defined and

$$\|\nabla(u - I_{\mathscr{T}} u)\|_{L^2(T)} \leq C\ |u|_{W_1^2(T)} =: e_{\mathscr{T}}(u, T) \qquad \forall\, T \in \mathscr{T}.$$

- **Why the absence of h_T matters?** The estimate is written with equal Sobolev numbers

$$\mathrm{sob}(W_1^2) = 2 - \tfrac{2}{1} = 0 = 1 - \tfrac{2}{2} = \mathrm{sob}(H^1),$$

whence no positive power of h_T. This scale invariance makes the discrete optimization below clean.

- **Space $W_p^2(\Omega)$:** results below for $W_1^2(\Omega)$ are valid for $W_p^2(\Omega)$ with $p > 1$, because $W_p^2(\Omega) \hookrightarrow W_1^2(\Omega)$.

Discrete minimization problem

- **Total error bound:** Collect the surrogate local errors into $\mathrm{e} = (e_{\mathscr{T}}(u,T))_{T\in\mathscr{T}} \in \mathbb{R}^N$, $N = \#\mathscr{T}$, and consider the total (squared) error which accumulates in ℓ^2

$$E_{\mathscr{T}}(u)^2 := \sum_{T\in\mathscr{T}} e_{\mathscr{T}}(u,T)^2.$$

- L^1**-constraint:** Accumulating the surrogate errors in ℓ^1 leads to the global regularity $W_1^2(\Omega)$.

$$\sum_{T\in\mathscr{T}} e_{\mathscr{T}}(u,T) = C\, |u|_{W_1^2(\Omega)}\,.$$

The ℓ^1-summability reflects regularity $W_1^2(\Omega)$, i.e. the equality $\mathrm{sob}(W_1^2) = \mathrm{sob}(H^1)$.

- **Question:** Among all ways of distributing the fixed quantity $\sum_{T\in\mathscr{T}} e_{\mathscr{T}}(u,T)$ over N elements, which minimizes $E_{\mathscr{T}}(u)^2$?

Equidistribution is optimal

- **Lagrangian:** Idealize $e_{\mathscr{T}}(u, T)$ to range over all nonnegative reals and form the Lagrangian

$$\mathscr{L}[\mathrm{e}, \lambda] = \sum_{T \in \mathscr{T}} e_{\mathscr{T}}(u, T)^2 - \lambda \Big(\sum_{T \in \mathscr{T}} e_{\mathscr{T}}(u, T) - C \, |u|_{W^2_1(\Omega)} \Big).$$

- **Optimality condition:** The condition $\partial \mathscr{L} / \partial e_{\mathscr{T}}(u, T) = 2 \, e_{\mathscr{T}}(u, T) - \lambda = 0$ forces

$$e_{\mathscr{T}}(u, T) = \frac{\lambda}{2} \qquad \forall \, T \in \mathscr{T} :$$

the local error is the same on every element.

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  the local error is the same on every element.

- **Conclusion:**

A graded mesh is quasi-optimal *if the local error is equidistributed.*

The resulting rate

- **Eliminating λ:** Equidistribution $e_{\mathscr{T}}(u,T) = \frac{\lambda}{2}$ gives

$$E_{\mathscr{T}}(u)^2 = \frac{\lambda^2}{4} N, \qquad C\,|u|_{W_1^2(\Omega)} = \frac{\lambda}{2} N.$$

Eliminating λ yields the optimal decay rate

$$\boxed{\|\nabla(u - I_{\mathscr{T}} u)\|_{L^2(\Omega)} \le C\,|u|_{W_1^2(\Omega)}\, N^{-1/2} \le C\,|\Omega|^{1/q}\,|u|_{W_p^2(\Omega)}\, N^{-1/2},}$$

where $q = \frac{p}{p-1}$ for $p > 1$.

- **Interpretation:**
  - **Optimal rate:** Same rate $N^{-1/2}$ as *uniform* refinement for $u \in H^2(\Omega)$;
  - **Weaker regularity:** It requires only $u \in W_p^2(\Omega)$, with $p \ge 1$, a much larger space.
  - **Nonlinear approximation:** $\mathscr{T}$ is designed for the given u.

Example: W^2_p-regularity at a reentrant corner

- **Corner singularity:** Let $u(r,\theta) \sim r^\gamma$, $0 < \gamma < 1$ — the typical corner singularity. Then $D^2 u \sim r^{\gamma-2} \notin L^2$, so $u \notin H^2(\Omega)$; but

$$\|D^2 u\|^p_{L^p(\Omega)} = c \int_0^1 r^{p(\gamma-2)}\, r\,\mathrm{d}r = c \int_0^1 r^{p(\gamma-2)+1}\,\mathrm{d}r < \infty \quad \Longrightarrow \quad u \in W^2_p(\Omega),$$

provided $1 \le p < p_0 = \frac{2}{2-\gamma}$.

- **Optimal rate via grading:** Hence $|u|_{W^2_p(\Omega)} < \infty$, and a mesh that equidistributes the local energy error — necessarily graded toward the corner — achieves

$$\|\nabla(u - I_{\mathscr{T}} u)\|_{L^2(\Omega)} \lesssim |u|_{W^2_p(\Omega)}\, N^{-1/2},$$

the optimal 2-D rate *despite* $u \notin H^2(\Omega)$.

Fractional regularity and the general rate

- **Fractional smoothness:** What dictates the exponent? For piecewise linears ($k = 1$) approximation theory allows fractional $1 < s \leq 2$. The local estimate persists,

$$\|\nabla(u - I_{\mathscr{T}} u)\|_{L^2(T)} \leq C \, |u|_{W_p^s(\omega_T)} = e_{\mathscr{T}}(u, T) \quad \forall\, T \in \mathscr{T}, \qquad \mathrm{sob}(W_p^s) - \mathrm{sob}(H^1) = s - \frac{2}{p} = 0,$$

with $I_{\mathscr{T}}$ a quasi-interpolant and ω_T a discrete neighborhood.

- **Accumulating in ℓ^p and ℓ^2:** $e_{\mathscr{T}}(u, T) = \Lambda$ (constant) yields

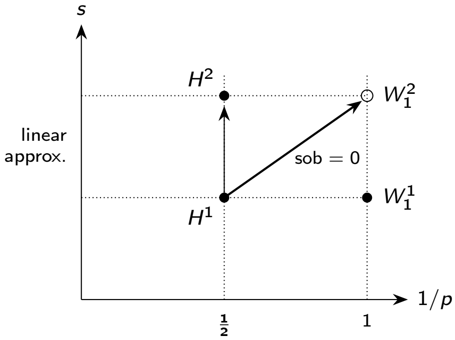
$$\sum_{T \in \mathscr{T}} e_{\mathscr{T}}(u, T)^p = \Lambda^p N \approx |u|_{W_p^s(\Omega)}^p, \qquad E_{\mathscr{T}}(u)^2 = \Lambda^2 N.$$

- **Outcome:** $\Lambda \leq C \, |u|_{W_p^s(\Omega)} N^{-1/p}$, and altogether

$$\boxed{\|\nabla(u - I_{\mathscr{T}} u)\|_{L^2(\Omega)} \leq C \, |u|_{W_p^s(\Omega)} \, N^{-\frac{s-1}{2}}.}$$

The numerator $s - 1$ is the difference of differentiability indices between W_p^s and H^1.

The DeVore diagram



The oblique Sobolev line $\mathrm{sob}(W_p^s) = s - \frac{2}{p} = 0$ (that of H^1) governs nonlinear approximation.

- **Linear** theory raises s from 1 to 2 with $p = 2$: reaches H^2.
- **Nonlinear** theory trades integrability ($p = 2 \to 1$) for differentiability ($s = 1 \to 2$): reaches W_1^2.

Both attain the optimal rate $N^{-(s-1)/2}$: graded meshes with regularity W_p^s while uniform meshes with regularity H^s.

Practical question:

- How to construct graded meshes automatically and locally that equidistribute the error, while preserving conformity and shape regularity?
- The answer is bisection.

The Bisection Method

Newest-vertex bisection

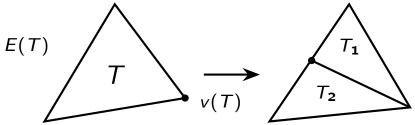
- **The refinement engine:** The refinement engine rests on a single operation: bisection of one element. Each T carries a distinguished refinement edge $E(T)$ and opposite a vertex $v(T)$.

Bisecting T joins $v(T)$ to the midpoint of $E(T)$, producing children T_1, T_2 sharing the newly created vertex $v(T_1) = v(T_2)$; each child gets its own refinement edge opposite the new vertex.

The *bisection method* involves two ingredients:

- a labeling of $\mathscr{T}_0$ fixing each element's refinement edge;
- a rule propagating labels to the children.

For $d = 2$ this is *newest-vertex bisection*; for $d > 2$ one adds element *type* and *vertex order*.



$T \to T_1, T_2$; $v(T_1) = v(T_2)$ is the new vertex.

Generation and labeling

- **Generation:** The generation $g(T)$ is the number of bisections needed to create T from its ancestor $T_0 \in \mathscr{T}_0$. Since each bisection halves area and $h_T = |T|^{1/2}$,

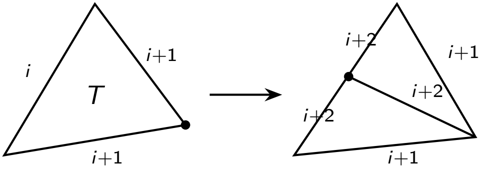
$$\boxed{h_T = 2^{-g(T)/2}\, h_{T_0}, \qquad T_0 \in \mathscr{T}_0 \text{ the ancestor of } T.}$$

- **Labeling:** Assign the ordered edge label $(i, i+1, i+1)$ to each T of generation $g(T) = i$, the level-i edge being $E(T)$.

Bisection splits the level-i edge and labels both new half-edges *and* the bisector $i+2$, leaving the rest unchanged:

$$(i, i{+}1, i{+}1) \longrightarrow (i{+}1, i{+}2, i{+}2)$$

for both children; the level-$(i+1)$ edges are the new refinement edges.



Shape regularity

- **Bisection preserves shape:** bisection never degrades element shape. The labeling prevents consecutive bisections of the same edge, inducing a rotation of refinement edges – a key geometric property.

Lemma 1 (shape regularity)

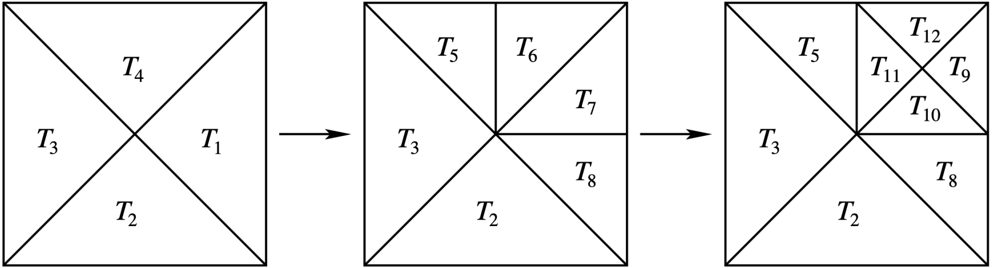
Every partition generated from $\mathscr{T}_0$ by newest-vertex bisection satisfies a uniform minimal-angle condition: the ratio of element diameter to inscribed-ball diameter is bounded by a constant depending only on $\mathscr{T}_0$. Equivalently, the descendants of each $T_0 \in \mathscr{T}_0$ fall into finitely many similarity classes.

- **The refinement order:** Write $\mathbb{T}$ for the conforming refinements of $\mathscr{T}_0$ reachable by bisection, and $\mathscr{T}_* \geq \mathscr{T}$ when $\mathscr{T}_*$ refines $\mathscr{T}$. This partial order $(\mathbb{T}, \geq)$ is the arena of the theory: the loop (SEMR) SOLVE $\rightarrow$ ESTIMATE $\rightarrow$ MARK $\rightarrow$ REFINE, discussed in Lecture 6, produces

$$\mathscr{T}_0 \leq \mathscr{T}_1 \leq \cdots \leq \mathscr{T}_k \leq \cdots ,$$

and shape regularity keeps all interpolation and a-posteriori constants uniform along the way.

A sequence of bisection meshes



Meshes $\{\mathscr{T}_k\}_{k=0}^2$ from $\mathscr{T}_0 = \{T_i\}_{i=1}^4$ with longest edges labeled for bisection.
$\mathscr{T}_1$ arises by bisecting T_1, T_4; $\mathscr{T}_2$ by refining T_6, T_7. Every intermediate mesh is conforming.

Compatible labeling

- **Compatible initial labeling:** For refinement to terminate and stay conforming, an element and its neighbor must agree on the shared refinement edge. In 2-D this is secured by a *compatible initial labeling* of $\mathscr{T}_0$: label every edge 0 or 1 so that

  every $T \in \mathscr{T}_0$ has exactly two edges labeled 1 and one labeled 0,

  the label-0 edge being its refinement edge (the generation-0 instance of $(i, i+1, i+1)$).

Compatible patch

If two elements share their refinement edge they form a compatible patch. When bisected *together*, inserting one vertex and creating *no hanging node*, the refinement remains local to the patch.

- **Always arrangeable:** Such a labeling is not automatic but always arrangeable after refining $\mathscr{T}_0$: For $d = 2$ bisect every element of $\mathscr{T}_0$ twice and label the two newest edges 0, the rest 1. For $d > 2$ the analogous type/vertex-order condition is more delicate but still available.

Complexity of Bisection

The Refine module and completion chains

- **Module REFINE:** Given $\mathscr{T} \in \mathbb{T}$ and a marked set $\mathscr{M} \subset \mathscr{T}$, the module $\mathscr{T}_* = \mathrm{REFINE}\,(\mathscr{T}, \mathscr{M})$ returns the smallest conforming refinement in which every $T \in \mathscr{M}$ has been bisected at least once.
- **Hanging node:** Bisecting a marked T across $E(T)$ generally leaves a hanging node on the neighbor across that edge.
- **Restoring conformity:** Conformity is restored by bisecting the neighbor too — which may create a new hanging node, and so on.
- **Completion chain:** The completion runs along a chain of elements linked by refinement edges, each bisected in turn until a compatible patch is reached (terminating at $\partial\Omega$ or a compatible pair). Crucially, the chain can be as long as the generation $g(T)$ of the marked element.

Recursive refinement is nonlocal

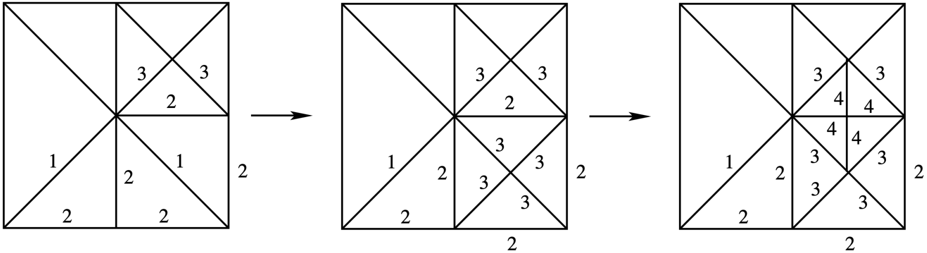


Figure: **Refinment chain:** Refining T_{10} forces the chain $\{T_{10}, T_8, T_2\}$, resolved in reverse: first T_2 (refinement edge on $\partial\Omega$, a compatible patch on its own), then T_8, finally T_{10}. The chain length is $g(T_{10}) = 3$, and it grows without bound as the mesh is refined.

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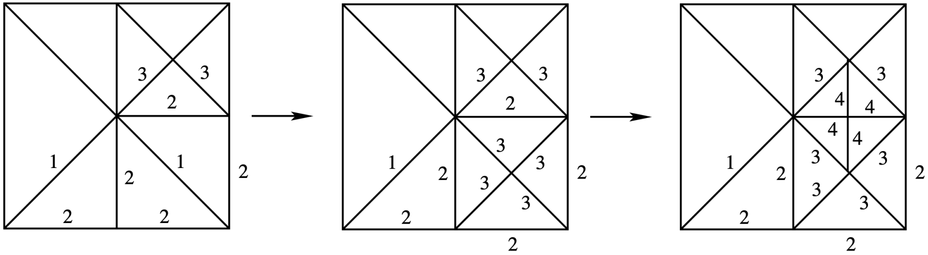

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Question: How can we related the cardinality $\#\mathscr{M}$ of $\mathscr{M}$ with $\#\mathscr{T}_*$?

Complexity of REFINE: a naive bound

- **Nonlocal process:** Because one mark can trigger a completion chain, the tempting *local* bound

$$\#\mathscr{T}_* - \#\mathscr{T} \;\leq\; \Lambda_0 \, \#\mathscr{M}$$

does not hold with a Λ_0 independent of the refinement level: the local cost may be as large as $g(T)$ for $T \in \mathscr{M}$, which grows without bound.

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- **The remedy:**

Do not bound a single step but the cumulative cost over the whole refinement history. The completion process is in average local and the total number of marks controls it.

- **Delicate property:** This is due to Binev–Dahmen–DeVore ($d = 2$) and Stevenson ($d \geq 3$).

Complexity of REFINE: Rigorous bound

Theorem 1 (complexity of bisection)

Suppose $\mathscr{T}_0$ satisfies the compatible labeling for $d = 2$ (the analogous type/vertex-order condition of Stevenson for $d > 2$). Let $\mathscr{M}_j \subset \mathscr{T}_j$ be the marked set at step j and $\mathscr{T}_{j+1} = \mathit{REFINE}\,(\mathscr{T}_j, \mathscr{M}_j)$. Then there is a constant Λ_0, depending only on $\mathscr{T}_0$ and d, such that for all $k \geq 1$

$$\boxed{\#\mathscr{T}_k - \#\mathscr{T}_0 \;\leq\; \Lambda_0 \sum_{j=0}^{k-1} \#\mathscr{M}_j.}$$

If each marked element is bisected a fixed number $b \geq 1$ of times (e.g. $b = 2$), the bound persists with Λ_0 also depending on b.

- **Key idea:** The above bound relies on the construction of an allocation function $\lambda(T, T_*)$ with suitable properties, as discussed next.

Heuristics: Allocation Function

- Each element $T_* \in \mathscr{M}$ is assigned a fixed amount C_1 of money to spend on refined elements in $\mathscr{T}$.
- Let $\lambda(T, T_*)$ be the portion of money spent by T_* on T. Then it must hold

$$\sum_{T \in \mathscr{T} \setminus \mathscr{T}_0} \lambda(T, T_*) \leq C_1 \qquad \forall T_* \in \mathscr{M} = \cup_{j=0}^{k-1} \mathscr{M}_j.$$

- The investment of all elements in $\mathscr{M}$ is fair in the sense that each $T \in \mathscr{T} \setminus \mathscr{T}_0$ gets at least a fixed amount C_2, whence

$$\sum_{T_* \in \mathscr{M}} \lambda(T, T_*) \geq C_2 \qquad \forall T \in \mathscr{T} \setminus \mathscr{T}_0.$$

- Summing up yields

$$C_2(\#\mathscr{T} - \#\mathscr{T}_0) \leq \sum_{T \in \mathscr{T} \setminus \mathscr{T}_0} \sum_{T_* \in \mathscr{M}} \lambda(T, T_*) = \sum_{T_* \in \mathscr{M}} \sum_{T \in \mathscr{T} \setminus \mathscr{T}_0} \lambda(T, T_*) \leq C_1 \, \#\mathscr{M}.$$

Why the bound matters

- **Combinatorial backbone of optimality:**

The total number of elements $\#\mathscr{T}_k - \#\mathscr{T}_0$ created by REFINE at step k is controlled by the total number of marked elements $\sum_{j=0}^{k-1} \#\mathscr{M}_j$ from step 1 to step k:

$$\#\mathscr{T}_k - \#\mathscr{T}_0 \;\leq\; \Lambda_0 \sum_{j=0}^{k-1} \#\mathscr{M}_j.$$

- **The stakes:** Without it, the completion process — however local it looks element by element — could in principle overwhelm the savings of adaptivity.
- **The payoff:** With it, the cost of keeping meshes conforming is provably a fixed multiple of the useful refined elements $\sum_{j=0}^{k-1} \#\mathscr{M}_j$.
- **Looking ahead:** This is exactly the ingredient the optimality theory (Lecture 9) needs.

The Greedy Algorithm

Greedy adaptive algorithm

- **Goal:** Given $u \in H_0^1(\Omega) \cap W_p^2(\Omega)$, $p > 1$, construct a near-optimal mesh by iterative bisection. Define the local energy error

$$e_{\mathscr{T}}(T) = e_{\mathscr{T}}(u, T) := \|\nabla(u - I_{\mathscr{T}} u)\|_{L^2(T)},$$

and let $\delta > 0$ be an error tolerance.

GREEDY $(\mathscr{T}_0, \delta)$

$\mathit{function}\ \mathit{GREEDY}(\mathscr{T}_0, \delta)$
$\mathscr{T} := \mathscr{T}_0$
while $\mathscr{M} := \{T \in \mathscr{T} : e_{\mathscr{T}}(T) > \delta\} \neq \emptyset$**:**
  $\mathscr{T} \leftarrow \mathit{REFINE}\,(\mathscr{T}, \mathscr{M})$
return $\mathscr{T}$

This assumes we know u and can evaluate $e_{\mathscr{T}}(T)$: a theoretical benchmark for what adaptive methods can achieve. In practice $e_{\mathscr{T}}(T)$ is replaced by a computable estimator (Lecture 5).

Optimal greedy approximation

Theorem 2 (optimal greedy approximation)

If $u \in H_0^1(\Omega) \cap W_p^2(\Omega)$ for some $p > 1$ and $d = 2$, the output $\mathscr{T}$ of GREEDY $(\mathscr{T}_0, \delta)$ satisfies

1. $\|\nabla(u - I_{\mathscr{T}} u)\|_{L^2(\Omega)} \le \delta\,(\#\mathscr{T})^{1/2}$;
2. $\#\mathscr{T} - \#\mathscr{T}_0 \lesssim \delta^{-1}\,|\Omega|^{1-1/p}\,|u|_{W_p^2(\Omega)}$.

In particular, with $N = \#\mathscr{T} \ge 2\#\mathscr{T}_0$,

$$\boxed{\|\nabla(u - I_{\mathscr{T}} u)\|_{L^2(\Omega)} \lesssim |\Omega|^{1-1/p}\,|u|_{W_p^2(\Omega)}\; N^{-1/2}.}$$

- **Interpretation:** $N^{-1/2}$ is the optimal rate for $d = 2$ but it is achieved with regularity $W_p^2(\Omega)$ with $p > 1$ rather than the limiting space $W_1^2(\Omega)$. This is due to the completion process of REFINE to keep conformity. We prove this theorem in five steps.

Proof — Steps 1 & 2

- **Step 1 — Local error bound (termination + statement (1)):** By the local interpolation estimate $e_{\mathscr{T}}(T) \le C\, h_T^r\, |u|_{W_p^2(T)}$ with

$$r = \mathrm{sob}(W_p^2) - \mathrm{sob}(H^1) = 2(1 - 1/p) > 0,$$

refining strictly decreases $e_{\mathscr{T}}(T)$ and the algorithm terminates. At termination $e_{\mathscr{T}}(T) \le \delta$ for all $T \in \mathscr{T}$, thereby giving

$$\|\nabla(u - I_{\mathscr{T}} u)\|_{L^2(\Omega)}^2 = \sum_{T \in \mathscr{T}} e_{\mathscr{T}}(T)^2 \le \delta^2\, \#\mathscr{T},$$

which is statement (1).

- **Step 2 — Counting elements:** Let $\mathscr{M}$ be the total set of all elements ever marked. By the complexity theorem $\#\mathscr{T} - \#\mathscr{T}_0 \le \Lambda_0\, \#\mathscr{M}$, so it suffices to bound $\#\mathscr{M}$. Partition $\mathscr{M}$ by size

$$\mathscr{P}_j = \{T \in \mathscr{M} :\ 2^{-(j+1)} \le |T| < 2^{-j}\}, \quad j \ge 0, \qquad \#\mathscr{M} = \sum_{j \ge 0} \#\mathscr{P}_j.$$

Proof — Steps 3 & 4

- **Step 3 — Bounding $\#\mathscr{P}_j$ (geometry):** Elements of $\mathscr{P}_j$ are disjoint: if $T_1, T_2 \in \mathscr{P}_j$ had intersecting interiors then, by the bisection procedure, one contains the other, say $T_1 \subset T_2$, so $|T_1| \leq \frac{1}{2}|T_2|$ — contradicting the definition of $\mathscr{P}_j$. Since each has area $\geq 2^{-(j+1)}$,

$$2^{-(j+1)} \# \mathscr{P} \leq |\Omega| \quad \Rightarrow \quad \#\mathscr{P}_j \leq |\Omega|\, 2^{j+1}. \tag{geom}$$

- **Step 4 — Bounding $\#\mathscr{P}_j$ (regularity):** Every $T \in \mathscr{P}_j$ was marked because $e_{\mathscr{T}}(T) > \delta$, and $h_T = |T|^{1/2} \leq 2^{-j/2}$, so

$$\delta < e_{\mathscr{T}}(T) \lesssim h_T^r\, |u|_{W_p^2(T)} \lesssim 2^{-jr/2}\, |u|_{W_p^2(T)}\,.$$

Raising to the p-th power (to account for summability of $|u|_{W_p^2(T)}$) and summing over $T \in \mathscr{P}_j$,

$$\delta^p\, \#\mathscr{P}_j \lesssim 2^{-jrp/2}\, |u|^p_{W_p^2(\Omega)}\,. \tag{sob}$$

Proof — Step 5 (crossover and summation)

- **Crossover index:** Bounds (geom) and (sob) are complementary and cross at an index j_0 with

$$2^{j_0} \approx \delta^{-p} 2^{-j_0 \frac{rp}{2}} |u|_{W^2_p(\Omega)} \quad \Rightarrow \quad 2^{j_0} \approx \delta^{-1} \, |u|_{W^2_p(\Omega)} \, |\Omega|^{-1/p},$$

because $1 + \frac{rp}{2} = p$.

- **Summation:** Both geometric series, for $j \leq j_0$ and $j > j_0$, relate to 2^{j_0} as follows:

$$\sum_{j \leq j_0} 2^j \approx 2^{j_0}, \qquad \sum_{j > j_0} \left(2^{-rp/2}\right)^j \approx \left(2^{-rp/2}\right)^{j_0} = 2^{-(p-1)j_0}.$$

Therefore

$$\#\mathcal{M} \lesssim |\Omega| \, 2^{j_0} \approx \delta^{-1} \, |u|_{W^2_p(\Omega)} \, |\Omega|^{1-1/p}.$$

This is statement (2), and together with (1) completes the proof. □

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3. **Bisection algorithm:** newest-vertex bisection with compatible labeling builds graded, conforming, shape-regular meshes by local operations.
4. **Complexity (BDV/Stevenson):** $\#\mathscr{T}_k - \#\mathscr{T}_0 \leq \Lambda_0 \sum_j \#\mathscr{M}_j$ — total marks control total DOFs.

5. **Greedy optimality:** $u \in W^2_p$ $(p > 1)$ exhibits optimal decay rate $N^{-1/2}$, i.e. $u \in \mathscr{A}_{1/2}$ approximation class to be introduced in Lecture 9..

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Next: Lecture 5 — a posteriori error analysis (computable estimators replacing $e_{\mathscr{T}}(T)$).